Learning

Neural Networks

photo: fisher-price.com

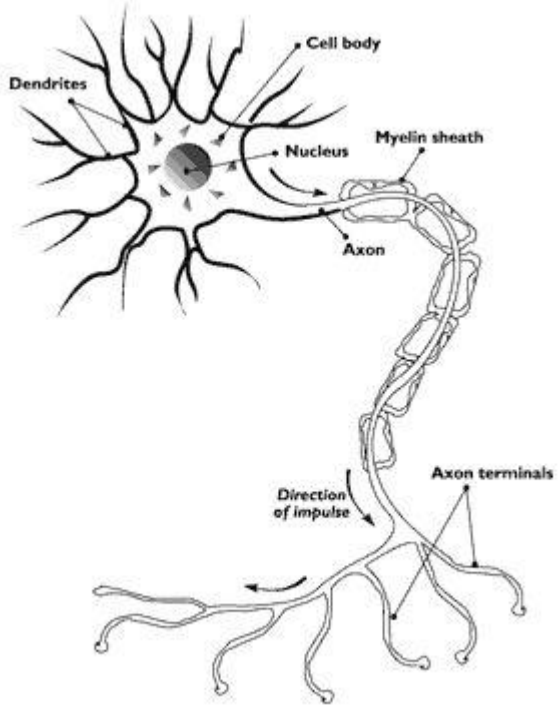
Xuanji Yu,

Courtesy of Professor Marie Roch

Chapter 19, Russell & Norvig

Connectionist networks (artificial neural networks)

Neuroscience—hypothesis that mental activity consists primarily of electrochemical activity in networks of brain cells called neurons

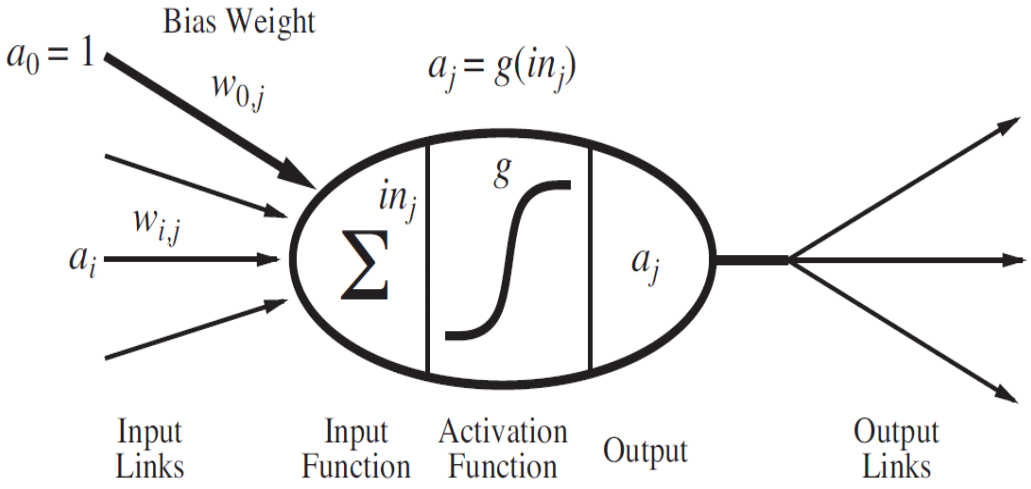


Neuron

National Institute on Drug Abuse

Roughly speaking, neuron "fires" when a linear combination of its inputs exceeds some (hard or soft) threshold—that is, it implements a linear classifier

A neural network is just a collection of units connected together



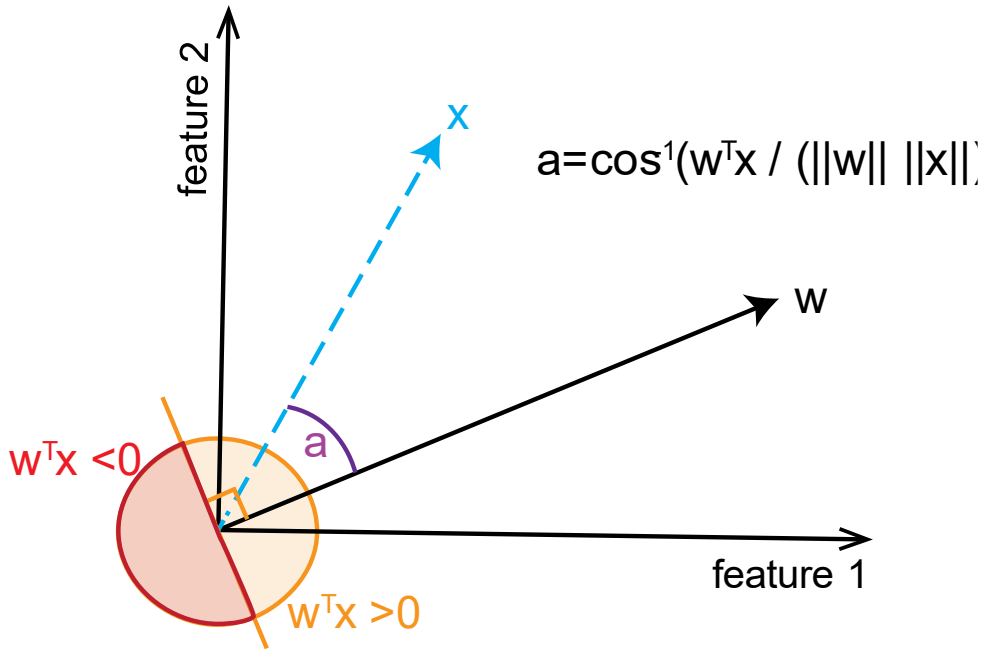
Model of a neuron

Fig. 18.19 R&N

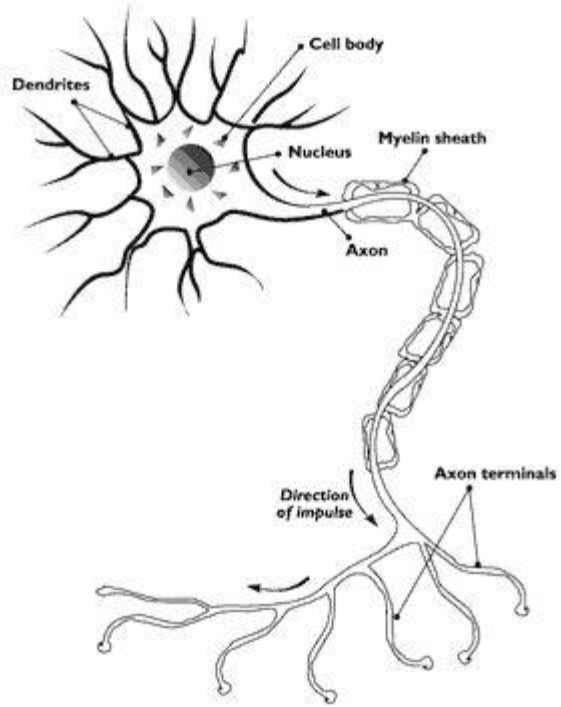
$$a = g(w^T x + b) = g\left(\sum_i w_i x_i + b\right)$$

Weight W determines strength and sign of the connection

Interpreting weight vectors



Roch et al. 2021, *Acoustics Today*

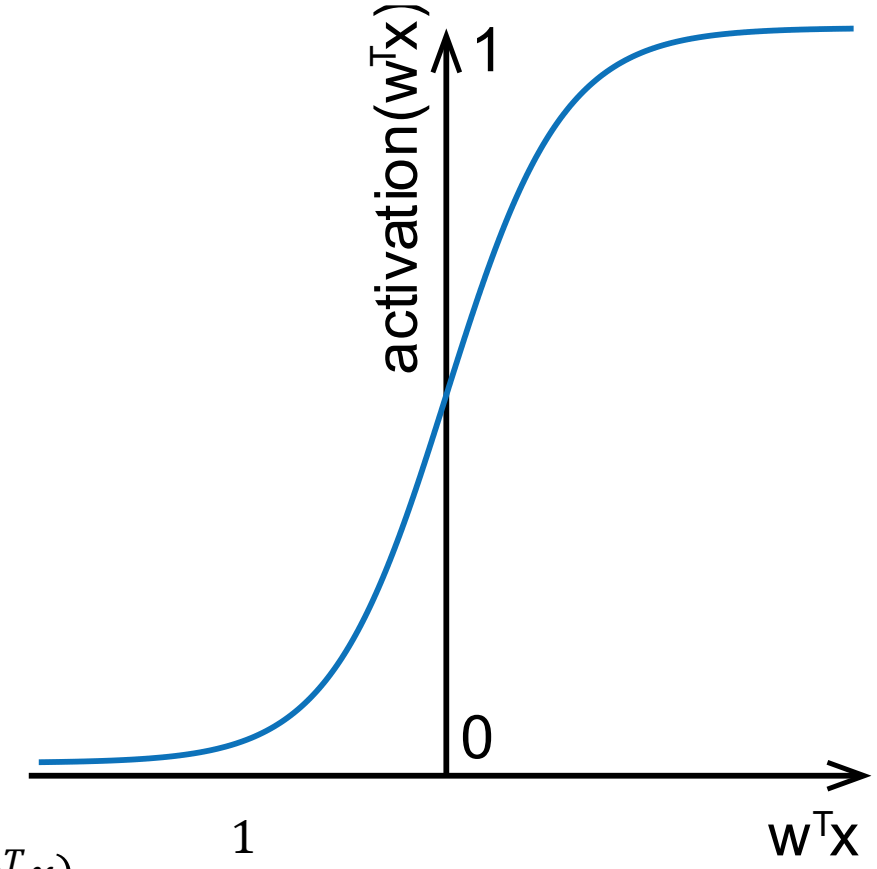


- $w^T x \propto \angle a$
  (note: $w^T x = \|w\| \cdot \|x\| \cdot \cos(a)$)
- Sign indicates which side of line $\perp$ to w vector x falls on

Activation function

$$a = g(w^T x + b) = g\left(\sum_i w_i x_i + b\right)$$

- The dot product is passed through an activation function.
- Key ideas about activation functions:
  - nonlinear
  - differentiable
- Common functions:
  - sigmoid or logistic regression (shown)
  - rectified linear unit (ReLU)



$$\sigma(w^T x) = \frac{1}{1 + e^{-w^T x}}$$

Roch et al. 2021, *Acoustics Today*

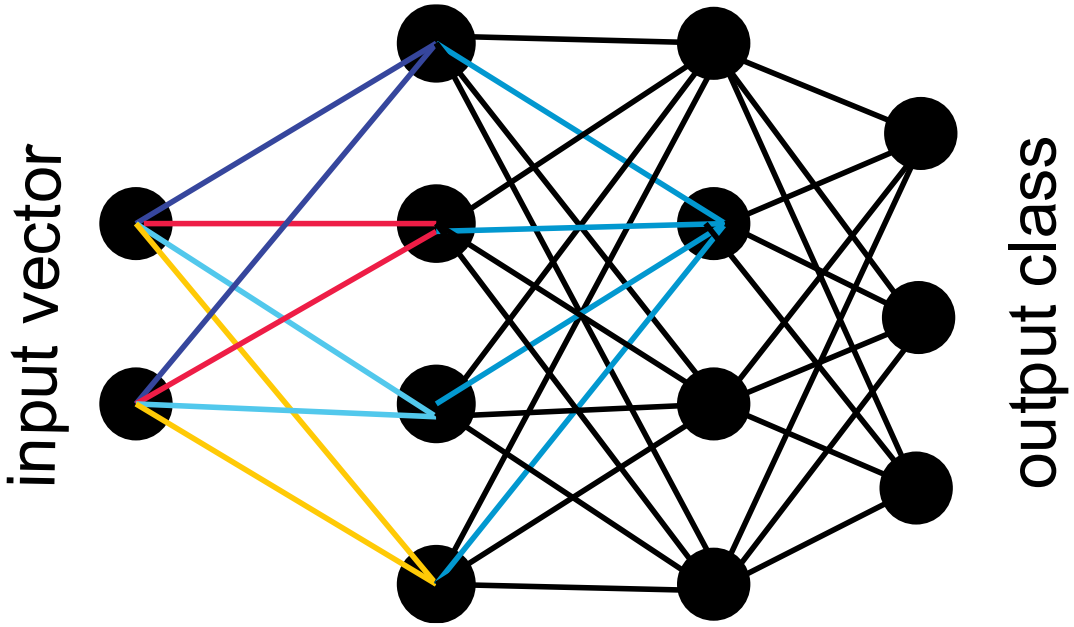
Connectionist networks

- Activations functions for perceptrons are nonlinear:
  - hard threshold
  - logistic regression (frequently called sigmoid function)
  - entire network of units can represent a nonlinear function

- Linking perceptrons together provides complex function modeling capability

Putting it together

- Feature vectors are presented to each node of the network
- Each node computes an output
- Subsequent nodes take previous inputs



derived from Roch et al. 2021, *Acoustics Today*

Output layer

- Output values can be regression targets
- Classification targets
  - probability of one class in a binary class problem
  - set of output vectors with probability of each class
- Labels for training
  - 1 in target category, “one-hot” representation
  - Example:
    - Original categorical variable:
      - Red
      - Green
      - Blue
    - One-hot encoded representation:
      - Red: [1, 0, 0]
      - Green: [0, 1, 0]
      - Blue: [0, 0, 1]

image: www.teacherspayteachers.com

P(cookie|image)

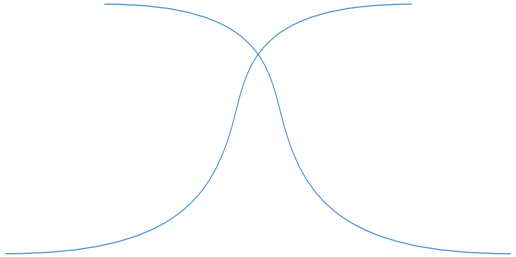
P(drum|image)

…

P(duck|image)

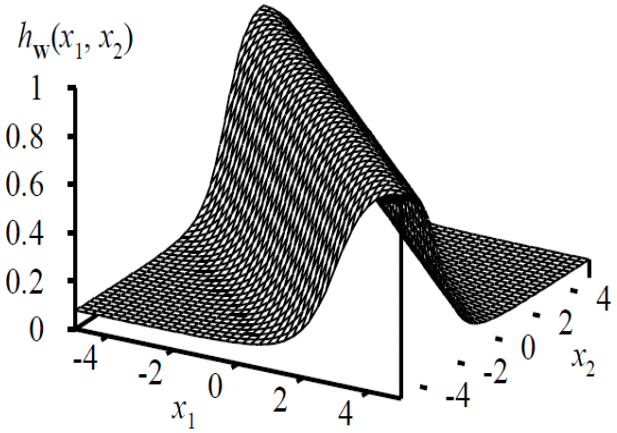
An intuitive view of neural nets

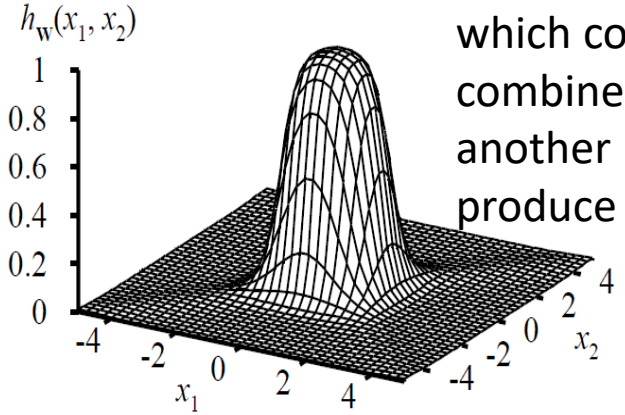
- Suppose we combine two perceptrons whose output functions are reversed



- This could be used to model a ridge in output space

vector function $h_w(x) = activation(w^T x)$





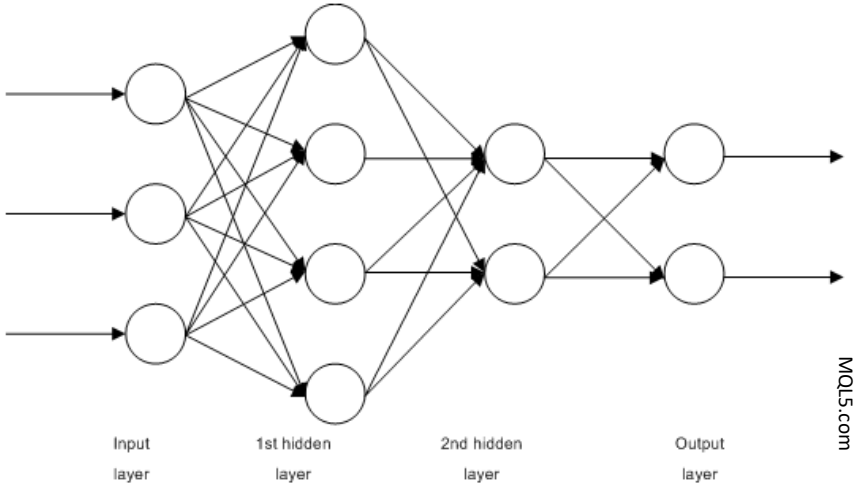
which could be combined with another ridge to produce this

Fig. 18.23 R&N

Learning in a neural network

- Consider input vector **x**
- Output vector **a**

$\vec{x}$



$\vec{a}$

Multilayer Feed-forward network

Learning in a neural network

- Similar to the regression problem, for output **a** and desired output **y**, we can find the loss gradient for each output node

$$\frac{\partial}{\partial w} Loss(w) = \frac{\partial}{\partial w} |y - h_w(x)|^2 = \frac{\partial}{\partial w} \sum_{k=1}^{D} (y_k - a_k)^2 = \sum_{k=1}^{D} \frac{\partial}{\partial w} (y_k - a_k)^2$$

$$h_w(x) = activation(w^T x)$$

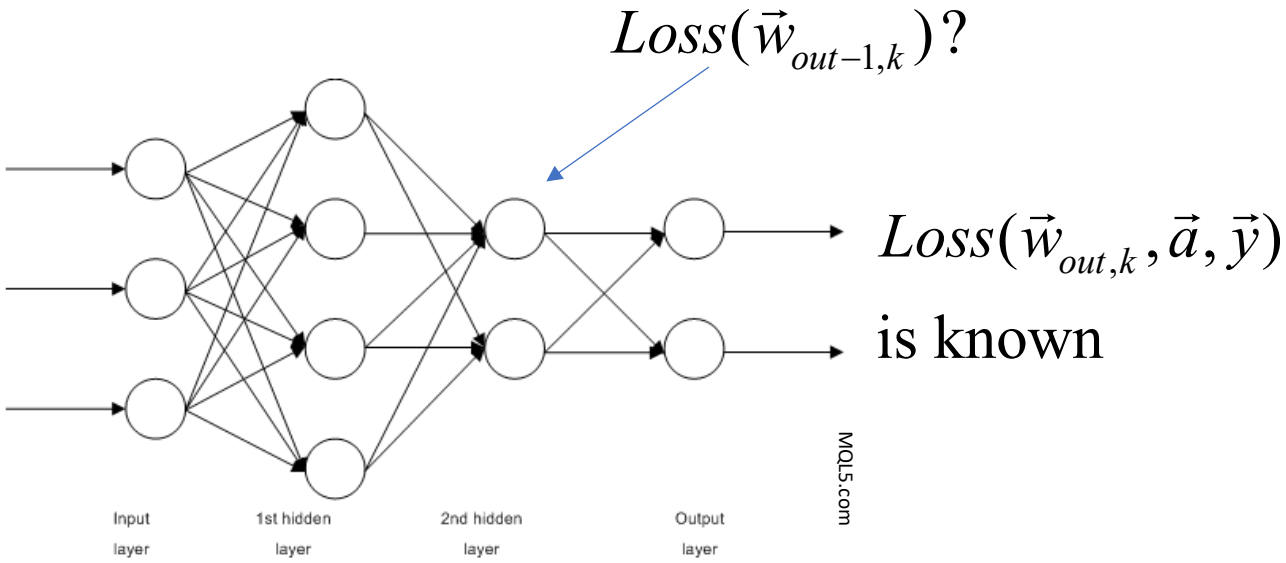
$$a_k = \frac{1}{1 + e^{-w_{output,k} \cdot input}}$$

here we assumed a sigmoid activation (other functions possible)

and use the perceptron learning rule for the sum of the gradients at the output layer.

Back-propagation

- What should the targets be for the previous input layer?



Multilayer Feed-forward network

Back propagating error (overview)

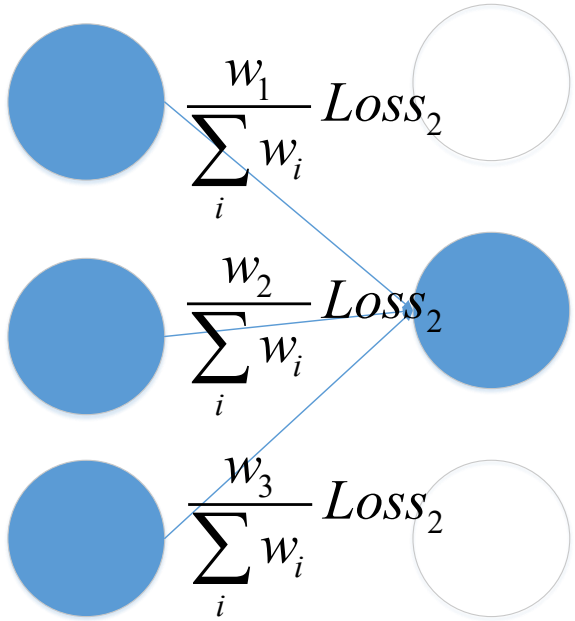
- Error of the kth output: $$Err_k = y_k - a_k$$

- We can compute the gradient for any input node (in) and apply the regression rule.

- This gives us a new set of weights for the output node.

Back propagating error (overview)

- After applying the update to the output layer, there still exists loss
- We assign a portion of the loss to each of the input nodes based on their weight.
- This contribution is computed for each node of the current layer

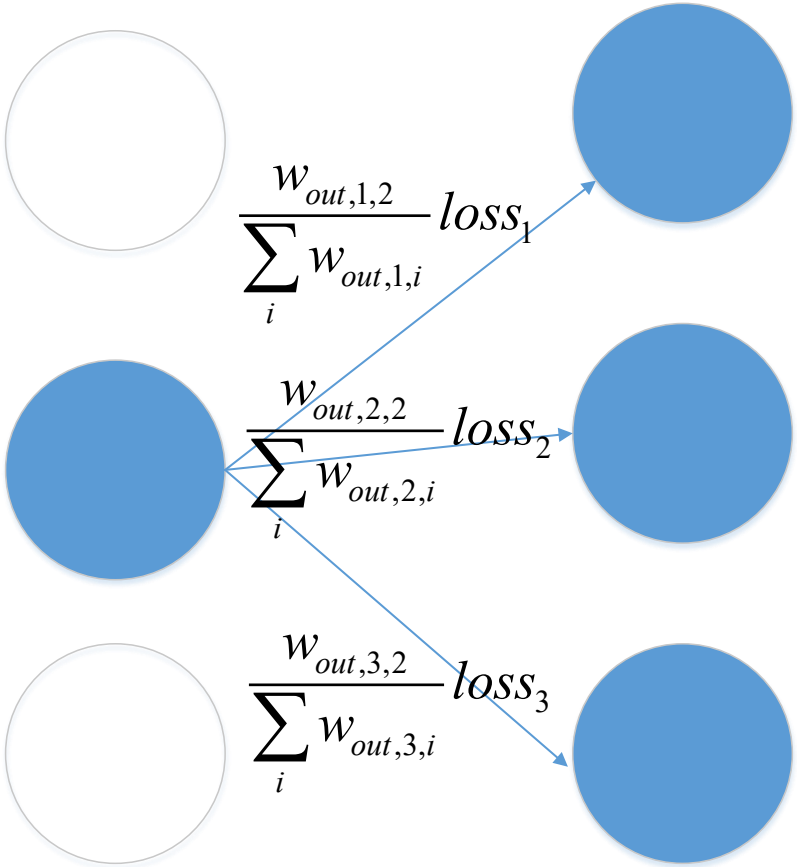


Back propagating error (overview)

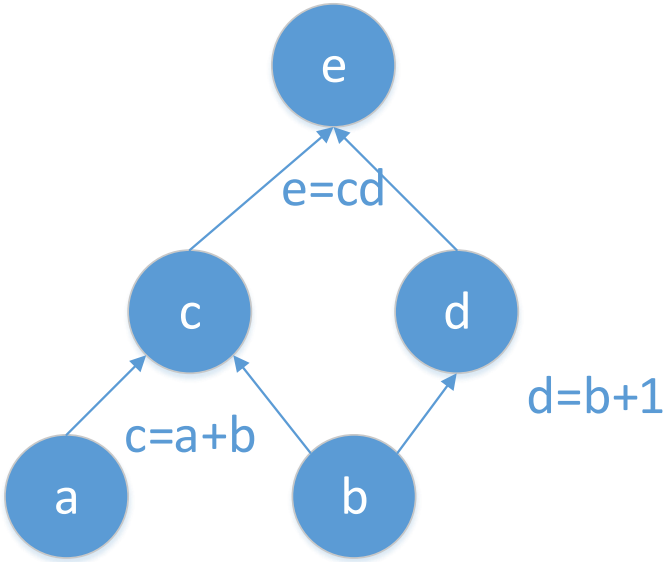
- Now we can look at the sum of losses attributable to each node in the previous layer.

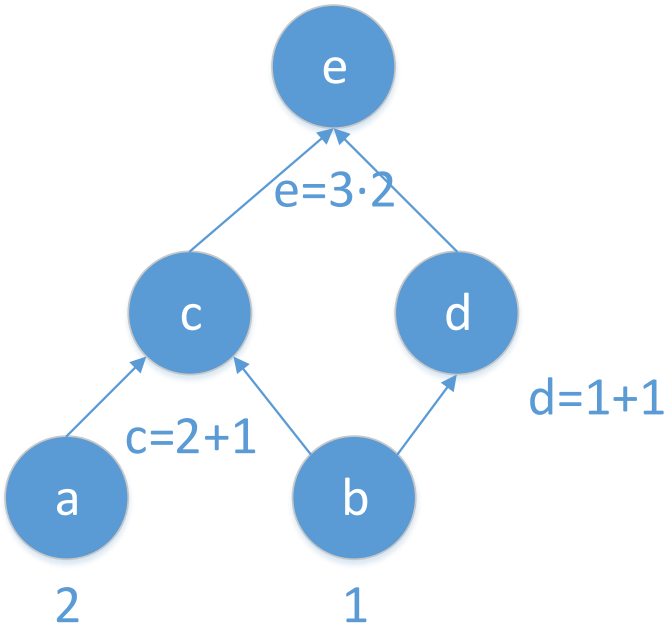
- The sum of these provides us with a loss to minimize.

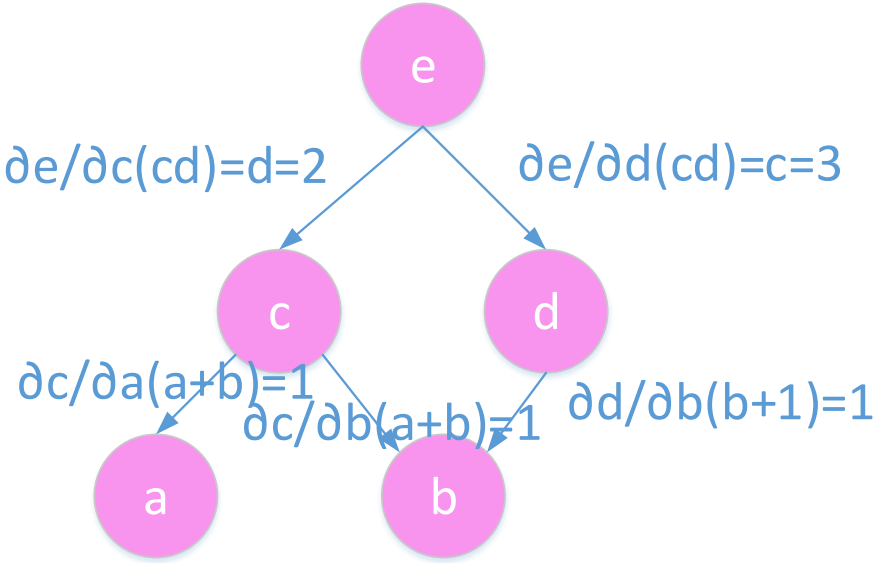
- Repeat recursively



$w^l_{j,k}$ layer l weight from node k to j







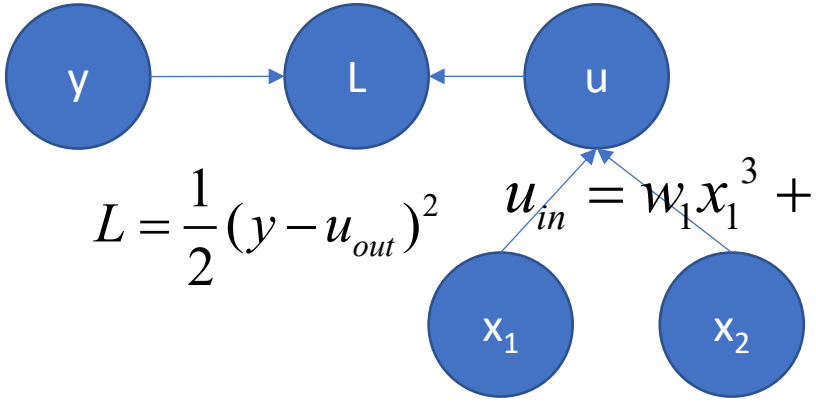
$$\frac{\partial e}{\partial b} = \frac{\partial e}{\partial c}\frac{\partial c}{\partial b} + \frac{\partial e}{\partial d}\frac{\partial d}{\partial b} = 2 \cdot 1 + 3 \cdot 1 = 5$$

Concrete example

Example based on
Christopher Olah's blog post

Activation fn example for backprop

$$u_{out} = \sigma(u_{in}) = \frac{1}{1+e^{-u_{in}}}$$



$$L = \frac{1}{2}(y - u_{out})^2 \qquad u_{in} = w_1 x_1^{\ 3} + w_2 x_2$$

partial derivatives

$$\frac{\partial L}{\partial u_{out}} = \frac{2}{2}(y - u_{out})(-1) = u_{out} - y$$

$$\frac{\partial u_{out}}{\partial u_{in}} = \sigma(u_{in})(1 - \sigma(u_{in}))$$

$$\frac{\partial u_{in}}{\partial x_1} = w_1 3 x_1^2$$

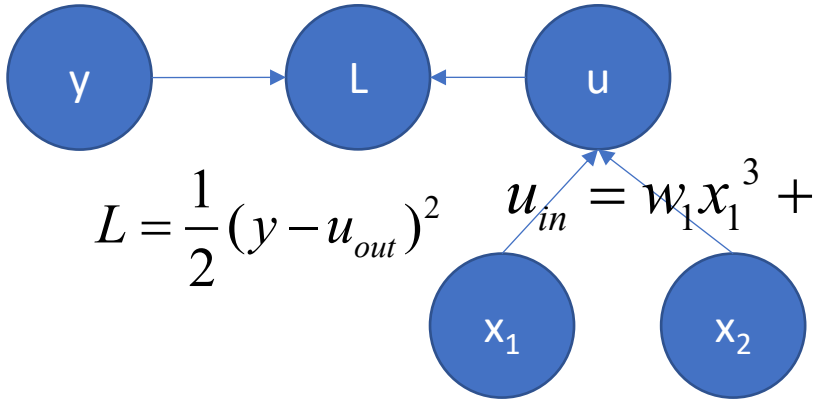
$$\frac{\partial u_{in}}{\partial u_{x_2}} = w_2$$

$$\frac{\partial u_{in}}{\partial w_1} = x_1^3$$

$$\frac{\partial u_{in}}{\partial w_2} = x_2$$

Activation fn example for backprop

$$u_{out} = \sigma(u_{in}) = \frac{1}{1+e^{-u_{in}}}$$



$$L = \frac{1}{2}(y - u_{out})^2 \qquad u_{in} = w_1 x_1^{\ 3} + w_2 x_2$$

To update w_1 we use the chain rule:

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial u_{out}} \frac{\partial u_{out}}{\partial u_{in}} \frac{\partial u_{in}}{\partial w_1} = (u_{out} - y)\sigma(u_{in})(1 - \sigma(u_{in}))x_1^3$$

from previous slide

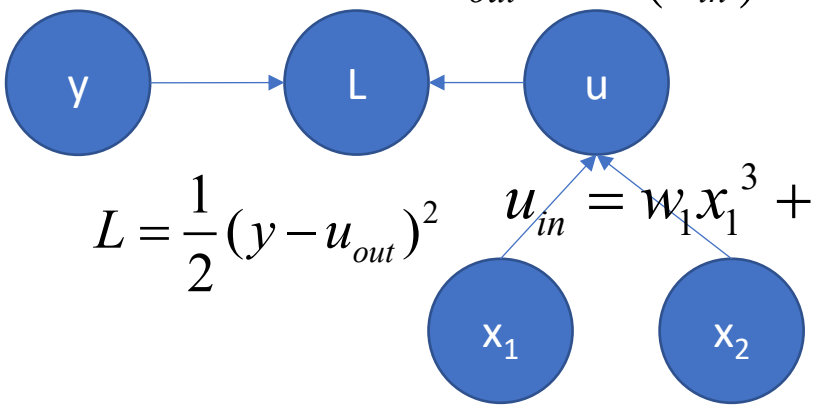
$$\frac{\partial L}{\partial u_{out}} = u_{out} - y$$

$$\frac{\partial u_{out}}{\partial u_{in}} = \sigma(u_{in})(1 - \sigma(u_{in}))$$

$$\frac{\partial u_{in}}{\partial w_1} = x_1^3$$

Activation fn example for backprop

$$u_{out} = \sigma(u_{in}) = \frac{1}{1+e^{-u_{in}}}$$



$$L = \frac{1}{2}(y - u_{out})^2 \qquad u_{in} = w_1 x_1^{\ 3} + w_2 x_2$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial u_{out}} \frac{\partial u_{out}}{\partial u_{in}} \frac{\partial u_{in}}{\partial w_1} = (u_{out} - y)\sigma(u_{in})(1 - \sigma(u_{in}))x_1^3$$

Concrete example

$$y = 0, w = \begin{bmatrix} .02 \\ .01 \end{bmatrix}, x = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

implies

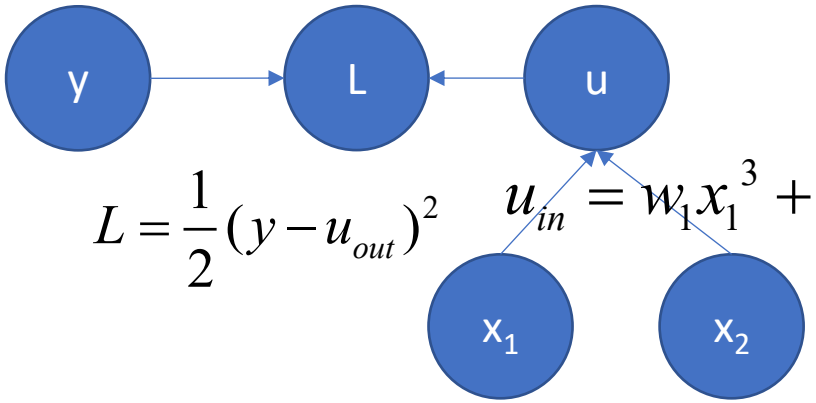
$$u_{in} = w_1 x^3 + w_2 x_2$$
$$= .02 \cdot 3^3 + .01 \cdot 5 = .59$$
$$u_{out} = \frac{1}{1+e^{-u_{in}}} = \frac{1}{1+e^{-.59}} = .6434$$
$$L = \frac{1}{2}(y - u_{out})^2$$
$$= .5 \cdot (0 - .6434)^2 = .2070$$

$$\frac{\partial L}{\partial u_{out}} = (u_{out} - y) \cdot -1 = .6434 - 0 = .6434$$
$$\frac{\partial u_{out}}{\partial u_{in}} = \sigma(u_{in})(1 - \sigma(u_{in})) = \sigma(.59)(1 - \sigma(.59))$$
$$= \frac{1}{1+e^{-.59}}\left(1 - \frac{1}{1+e^{-.59}}\right) = .6434(1 - .6434) = .2294$$
$$\frac{\partial u_{in}}{\partial w_1} = x_1^3 = 3^3 = 27$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial u_{out}} \frac{\partial u_{out}}{\partial u_{in}} \frac{\partial u_{in}}{\partial w_1} = .6434 \cdot .2294 \cdot 27 = 3.9851$$

$$u_{out} = \sigma(u_{in}) = \frac{1}{1+e^{-u_{in}}}$$

Activation fn example for backprop



$$L = \frac{1}{2}(y - u_{out})^2 \qquad u_{in} = w_1 x_1^{\ 3} + w_2 x_2$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial u_{out}} \frac{\partial u_{out}}{\partial u_{in}} \frac{\partial u_{in}}{\partial w_1} = (u_{out} - y)\sigma(u_{in})(1 - \sigma(u_{in}))x_1^3$$

Suppose we have a learning rate *ε=.01:* $w_1 = w_1 - \epsilon \frac{\partial L}{\partial w_1} = .02 - .01 \cdot 3.9851 = -0.0199$

Update of w_2 is left as an exercise, but loss with only w_1 changed:

$$y = 0, w = \begin{bmatrix} -.02 \\ .01 \end{bmatrix}, x = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

implies

$$u_{in} = w_1 x^3 + w_2 x_2 = -.02 \cdot 3^3 + .01 \cdot 5 = -0.49$$

$$u_{out} = \frac{1}{1 + e^{-u_{in}}} = \frac{1}{1 + e^{-(-0.49)}} = 0.3799$$

$$L = \frac{1}{2}(y - u_{out})^2 = .5 \cdot (0 - 0.3799\ )^2 = 0.0722 < \text{old L=.2070}$$

Overfit regressions

- Not a problem for univariate linear regression
- Problematic for multivariate
- Regularization provides penalties for increasing complexity

$$Cost(h_w) = EmpLoss(h) + \lambda Complexity(h)$$

- Common regularization: L_p penalties

$$Complexity(h_w) = L_p(w) = \sum_i |(w_i)|^p$$

- we regularize by picking the minimal cost hypothesis.

Loss

- Common loss functions

$$L_1(x, y, \hat{y}) = |y - \hat{y}|$$ absolute loss function

$$L_2(x, y, \hat{y}) = (y - \hat{y})^2$$ squared loss function

$$L_{0/1}(x, y, \hat{y}) = \begin{cases} 0 & y = \hat{y} \\ 1 & \text{otherwise} \end{cases}$$ 0/1 loss function

Regularization of regression

L_1 tends to produce sparse models with many zero weights.

L_2 tends to keep weights small overvall

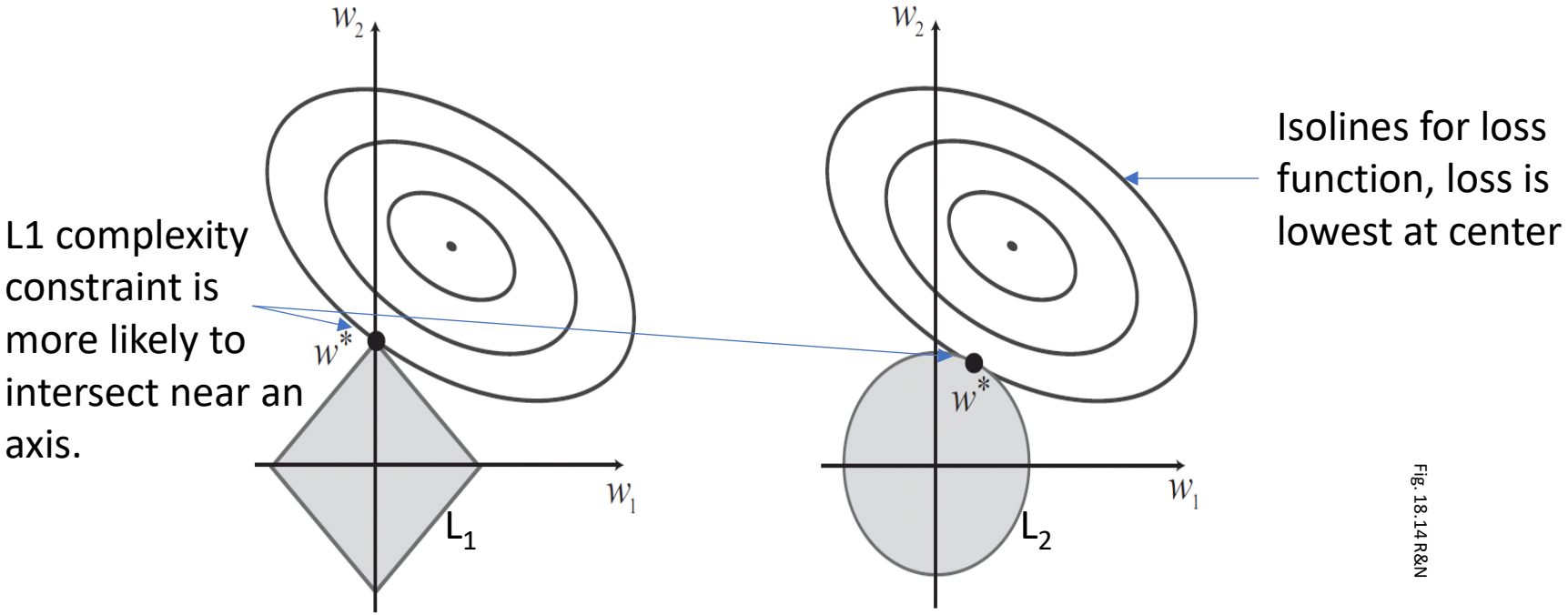


Fig. 18.14 R&N

- Minimizing Cost is equivalent to minimizing loss with constraint that complexity $\leq$ some constant.
- Complexity increases as w* moves away from the origin

Inputs

- Common to use a normalization on inputs
  - Learn the transform on the training data
  - Apply it to all data

- Commonly used transform
  - z-score normalization
  - Implemented in scikit learn's StandardScaler class

Outputs and loss functions

- Regression, commonly uses
  - sigmoid activation function
  - log mean squared error loss

- Classification
  - softmax activation on one-hot class outputs
  - cross-entropy loss

Neural net summary

- Supervised learner
  - Training labels either
    - High value for class (n classes → n output nodes)
    - Encoding of class information
    - Regression targets
  - Iterative training typically using a gradient descent algorithm (e.g. back propagation)
- Classification
  - Present features to input nodes
  - Interpret output nodes for category
- Caveats
  - Subject to overfit without appropriate regularization

Keras κέρας

- Library designed to simplify neural net specification
  - Originally designed to work with several neural net packages including Tensorflow
  - Now part of the official Tensorflow distribution
- Advantages (Simple)
  - High-level specification of neural nets and other computation.
  - Transparent GPU vs non-GPU programming
  - Rapid specification

http://vem.quantumunlimited.org/the-gates-of-horn/

Keras concepts : Models

Models can be:

- Specified: Functionality is specified by invoking model methods, e.g. add a new layer of N nodes.
- Compiled: A compile method writes the back-end code to generate the model
- Fitted: Optimization step where weights are learned
- Evaluated: Tested on new data

Keras concepts : Models

We can use a Sequential model for a feed-forward network

```
from tensorflow.keras.models import Sequential
model = Sequential()
```

Keras concepts: Layers

- Layers can be added to a model
- Dense layers
  - compute $f(W^T x+b)$
  - user specifies
    - number of units
    - input/output tensor shapes
      (tensors are N-dimensional arrays)
    - activation functions
    - other options…

A Keras model

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, InputLayer

model = Sequential()

# Three category prediction with 2 hidden layers
# and 30 features, categorical output (3 categories)
model.add(InputLayer(shape=(30,)))   # Note (30,) is a tuple w/one element
model.add(Dense(10, activation='relu'))
model.add(Dense(10, activation='relu'))
# Output probability of each category
model.add(Dense(3, activation='softmax'))
```

code for Tensorflow 2+

```
# Create the computational graph
# Specify type of gradient descent, loss metric, and
# measurement metric
model.compile(optimizer = "Adam",
              loss = "categorical_crossentropy",
              metrics = ["accuracy"])

# Not needed: prints architecture summary
model.summary()

# We need examples and labels for supervised learning
# examples:  samples X features numpy.array
examples = get_features()  # you write this

# samples X 1 vector of our 3 categories
labels = get_labels()  # you write this
```

```
from tensorflow.keras.utils import to_categorical

# Our network uses a Multinoulli distribution to
# output one of three choices.  Our labels are scalars,
# we need to convert these to vectors:
# 0 -> [1 0 0], 1 -> [0 1 0], 2 -> [0 0 1]
# this is sometimes called a “one-hot” vector

onehotlabels = to_categorical(labels)

# train the model
# 10 passes (epochs) over data, mini-batch size 100 examples
model.fit(examples, labels, batch_size=100, epochs=10)
```

Using a trained model

- To predict outputs

```
results = model.predict(examples)
```

  - results is Nx3 probabilities
  - What is the following?
    - `np.argmax(results, axis=1)`

Using a trained model

- To evaluate performance

```
# Returns list of metrics
results = model.evaluate(test_examples, test_labels)

# model.metrics_names tells us what was measured
# here:  ['loss', 'categorical_accuracy’]

print(results[1])   # accuracy
# In some fields, it is common to report error: 1 - accuracy
```

Regularization in keras

L1/L2 regularization is available as classes in keras

```
from tensorflow.keras.layers import Dense
from tensorflow.keras import regularizers
layer = Dense(
    units=64,
    kernel_regularizer=regularizers.L2(0.001)
)
```

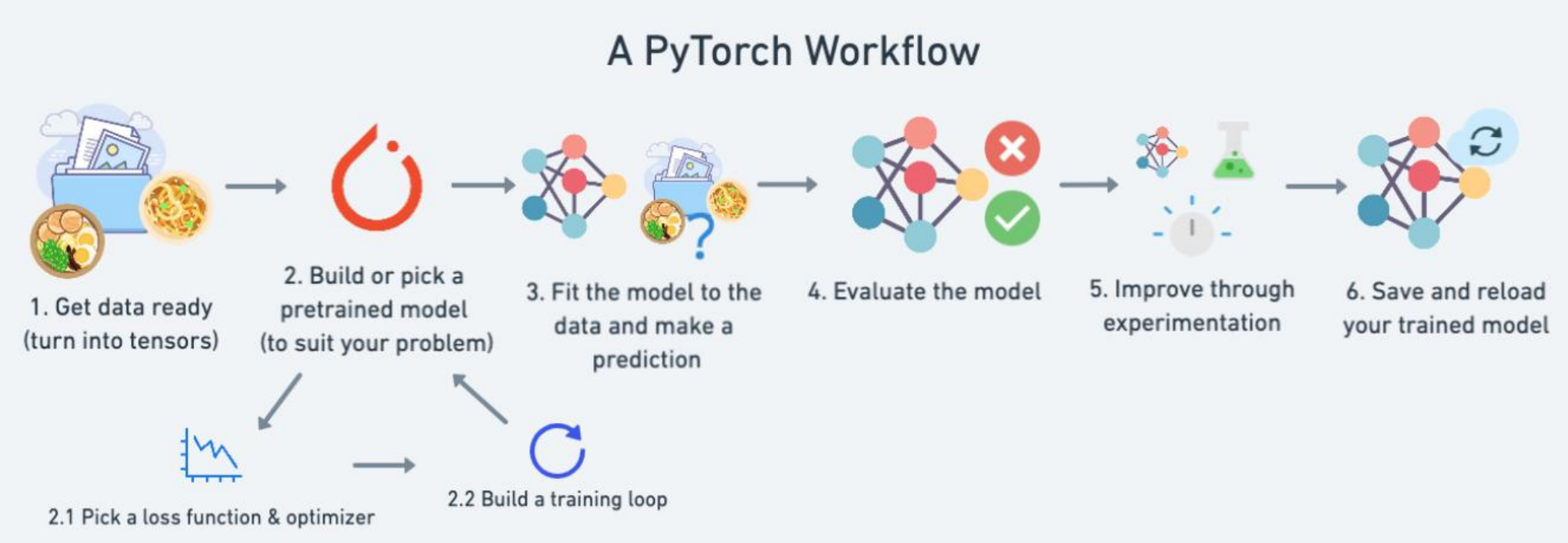
kernel regularizer regularizes the weights w (other types of regularizers are supported, but not used as often)

regularization factor = 0.001

$$Cost(h_w) = EmpLoss(h) + \lambda Complexity(h)$$

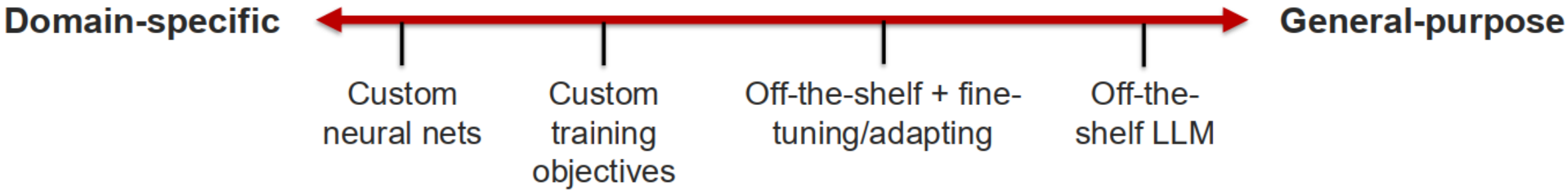
Pytorch

- Keras: Simple
- Pytorch: Full control, flexibility, research
  - import torch.nn, import torch.optim (Fundamentals)



HuggingFace

- "Huggingface" is a set of multiple packages
  - transformers: Provides API to initialize large pretrained models
  - datasets: Provides easy way to download datasets
  - pretrained models
    - Two General Modeling Paradigms



Neural net summary

- Disadvntaages
  - frequently hard to interpret
  - Many parameters require large data sets
  - Doesn’t do well with imbalanced examples
  - Slow to train
  - Overfits easily and regularization is important
- Advantages
  - Flexible, nonlinear learner
  - Deep architectures are very powerful

Useful links

https://playground.tensorflow.org/

https://distill.pub/

https://github.com/practical-tutorials/project-based-learning